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JAVA PROJECT REPORT ON

DDRMS Digital Document Record Management System

Submitted by

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DECLARATION

I hereby declare that the project entitled "**DDRMS DIGITAL DOCUMENT RECORD MANAGEMENT SYSTEM "** submitted in partial fulfillment of the requirements for the course **JAVA PROGRAMMING** is a **record of my original work** carried out under the guidance of , Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

THARANIISH R S

(Your Name)

CERTIFICATE

This is to certify that the project entitled **“DDRMS DIGITAL DOCUMENT RECORD MANAGEMENT SYSTEM ”** is a **bona fide work** carried out by **[THARANIISH R S] [I-BSc IT-.’A’]**, [], **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[Course Name]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date: 09.04.2026

Submitted for the Viva-Voce Examination held on **KG COLLAGE OF ARTS & SCIENCE**

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ABSTRACT

The **Document & Digital Record Management System (DDRMS)** is a Java-based application developed to efficiently manage and organize digital documents. In many organizations, records are often handled manually or stored in unstructured formats, leading to difficulties in searching, data duplication, loss of information, and inefficient record management.

This project aims to simplify document handling by providing an automated system that allows users to store, retrieve, update, and delete digital records easily. The system enables users to manage documents through a menu-driven interface, ensuring quick access and better organization of data.

The application is developed using Core Java programming concepts such as classes, objects, methods, loops, conditional statements, and file handling. It can be implemented as a console-based application or enhanced with a graphical user interface using Java Swing.

The main objective of the system is to improve efficiency, reduce manual effort, and ensure secure and structured storage of documents. The system provides features such as document entry, search functionality, record updating, and deletion, making it suitable for small organizations, offices, and educational institutions.

This project demonstrates the practical implementation of object-oriented programming in Java and provides a foundation for developing advanced document management systems with features like database integration, cloud storage, and enhanced security in the future.

1. INTRODUCTION

In today's digital era, managing documents efficiently is very important for organizations, institutions, and businesses. Traditionally, documents are stored in physical files or simple folders on computers. This manual method of document handling leads to several issues such as difficulty in searching files, data loss, duplication, and lack of proper organization.

With the growth of technology, there is a need for a computerized system that can store, manage, and retrieve documents efficiently. A Digital Record Management System helps users organize files systematically, ensuring easy access, security, and proper maintenance of records.

The **Document & Digital Record Management System (DDRMS)** is a Java-based application designed to manage documents digitally. The system allows users to add, view, search, update, and delete records efficiently. It ensures proper organization of files and improves productivity.

This project is developed using Core Java concepts such as classes, objects, methods, loops, conditional statements, and file handling. The system can be implemented as a console-based application or with a graphical interface using Java Swing.

The main goal of this system is to reduce manual work, improve data organization, and provide fast and secure access to digital records.

1.1 Objective

- The main objective of the DDRMS is to develop a simple and efficient system for managing digital documents using Core Java.
- The specific objectives are:
- • To design and develop a Java-based document management system
  • To store and manage digital records efficiently
  • To reduce manual effort in handling documents
  • To provide fast access and retrieval of files
  • To demonstrate Core Java concepts like classes, objects, loops, and file handling
  • To ensure proper organization and maintenance of records

1.2 Overview

The DDRMS is a menu-driven application developed using Java. It provides a platform where users can manage documents digitally.

The system allows users to:

• Add new documents
• View existing records
• Search documents by name or ID
• Update document details
• Delete records

The application organizes data efficiently and reduces the complexity of manual record management. It can be implemented as a console-based system or GUI using Java Swing.

This project demonstrates how Java can be used to develop real-world applications for data and document management

1.3 Features

he DDRMS includes the following features:

• **Add Document** – Allows users to store new documents
• **View Records** – Displays all stored documents
• **Search Function** – Finds documents quickly
• **Update Records** – Modify existing document details
• **Delete Records** – Remove unwanted files
• **User-Friendly Interface** – Simple menu-driven system
• **Efficient Storage** – Organizes records properly
• **Fast Access** – Quick retrieval of documents

• **Fast Billing Process** – Reduces the time required to generate customer bills.

2. PROGRAM ANALYSIS

Program analysis helps in understanding the problem and identifying system requirements.

The DDRMS focuses on solving problems related to manual document handling and improving efficiency.

2.1 Problem Definition

Manual document management creates several issues:

• Difficulty in searching documents
• Risk of data loss
• Duplicate records
• Time-consuming process
• Lack of proper organization

These problems reduce efficiency and productivity.

Therefore, a digital system is required to manage records efficiently.

The DDRMS provides a solution by storing and managing documents digitally.

2.2 Project Overview

The DDRMS is a Java-based system that simplifies document management.

The system provides:

• Adding new records
• Viewing stored documents
• Searching records
• Updating document details
• Deleting records

It uses Core Java concepts and can be implemented as console-based or GUI application.

2.3 Scopes and Limitations

Scope

• Managing digital records
• Organizing documents efficiently
• Fast retrieval of files

• Reducing manual errors
• Useful for offices, schools, and businesses

Limitations

• No database (if file-based system)
• Limited security features
• Not suitable for very large data systems
• Requires basic system knowledge

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The **DDRMS DIGITAL DOCUMENT RECORD SYSTEM** is designed as a menu-driven application where users can select food items, enter quantities, and generate a final bill automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

System design defines the structure and working of the system.

The DDRMS is divided into modules for better organization.

3.1 Modules

1. Document Entry Module

• Add new documents
• Store file details

2. View Module

• Display all stored records

3. Search Module

• Search documents by name or ID

4. Update Module

• Modify document details

5. Delete Module

• Remove records

6. Exit Module

• Close the application

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Hotel & Restaurant Billing System**, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

The DDRMS follows a structured development approach.

4.1 Problem Analysis

Identified issues:

• Manual record handling
• Time-consuming search
• Risk of data loss

Solution: Develop a digital system.

4.2 System Design

Modules designed:

• Add
• View
• Search
• Update
• Delete

4.3 Program Development

Implemented using Java:

• Classes and objects
• Menu-driven system
• File handling
• User input processing

4.4 Testing and Validation

Test cases used:

Test Case	Expected Result
Add record	Stored correctly
Search	Correct file displayed
Update	Data updated
Delete	Record removed

4.5 Result and Documentation

The system successfully:

• Stores documents
• Retrieves records
• Manages data efficiently

Documentation includes system design, code, and outputs.

4.6 Algorithm

1. Start program
2. Display menu
3. Select operation
4. Perform action (Add/Search/Update/Delete)
5. Repeat until exit
6. End program

4.7 Flowchart

Flow:

Start
↓
Display Menu
↓
Select Operation
↓
Process Request
↓
Display Result
↓
End

5. SOURCE CODE

There are three files are there there are

- Document.java
- DocumentManager.java
- Main.java

1. DOCUMENT.JAVA

```
public class Document {
    private int id;
    private String title;
    private String author;
    private String date;

    // Constructor
    public Document(int id, String title, String author, String date) {
        this.id = id;
        this.title = title;
        this.author = author;
        this.date = date;
    }

    // Getter
    public int getId() {
        return id;
    }
```

```
    // Convert object to string (for file saving)
    public String toString() {
        return id + "," + title + "," + author + "," + date;
    }

    // Convert string to object (for file loading)
    public static Document fromString(String data) {
        String[] parts = data.split(",");
        return new Document(
            Integer.parseInt(parts[0]),
            parts[1],
            parts[2],
            parts[3]
        );
    }

    // Display document
    public void display() {
        System.out.println("\nID: " + id);
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Date: " + date);
        System.out.println("------------------------");
    }
}
```

2.DOCUMENTMANAGER.JAVA

```
import java.io.*;
import java.util.*;
```

```
public class DocumentManager {

    private List<Document> documents;
    private final String FILE_NAME = "documents.txt";

    // Constructor
    public DocumentManager() {
        documents = new ArrayList<>();
        loadFromFile();
    }

    // Add document
    public void addDocument(Document doc) {
        documents.add(doc);
        saveToFile();
    }

    // View all documents
    public void viewDocuments() {
        if (documents.isEmpty()) {
            System.out.println("No documents found.");
            return;
        }

        for (Document doc : documents) {
            doc.display();
        }
    }
```

```
// Search document
public void searchDocument(int id) {
    for (Document doc : documents) {
        if (doc.getId() == id) {
            doc.display();
            return;
        }
    }
    System.out.println("Document not found.");
}

// Delete document
public void deleteDocument(int id) {
    boolean removed = documents.removeIf(doc -> doc.getId() == id);

    if (removed) {
        System.out.println("Document deleted.");
    } else {
        System.out.println("Document not found.");
    }

    saveToFile();
}

// Save to file
private void saveToFile() {
    try {
        BufferedWriter writer = new BufferedWriter(new FileWriter(FILE_NAME));
```

```
            for (Document doc : documents) {
                writer.write(doc.toString());
                writer.newLine();
            }

            writer.close();
        } catch (IOException e) {
            System.out.println("Error saving file.");
        }
    }

    // Load from file
    private void loadFromFile() {
        try {
            BufferedReader reader = new BufferedReader(new FileReader(FILE_NAME));
            String line;

            while ((line = reader.readLine()) != null) {
                documents.add(Document.fromString(line));
            }

            reader.close();
        } catch (IOException e) {
            System.out.println("No previous records found.");
        }
    }
}
```

3.MAIN.JAVA

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        DocumentManager manager = new DocumentManager();

        while (true) {
            System.out.println("\n===== DDRMS MENU =====");
            System.out.println("1. Add Document");
            System.out.println("2. View Documents");
            System.out.println("3. Search Document");
            System.out.println("4. Delete Document");
            System.out.println("5. Exit");
            System.out.print("Enter choice: ");

            int choice = sc.nextInt();

            switch (choice) {
                case 1:
                    System.out.print("Enter ID: ");
                    int id = sc.nextInt();
                    sc.nextLine();

                    System.out.print("Enter Title: ");
                    String title = sc.nextLine();
```

```
            System.out.print("Enter Author: ");
            String author = sc.nextLine();

            System.out.print("Enter Date: ");
            String date = sc.nextLine();

            manager.addDocument(new Document(id, title, author, date));
            break;

        case 2:
            manager.viewDocuments();
            break;

        case 3:
            System.out.print("Enter ID: ");
            manager.searchDocument(sc.nextInt());
            break;

        case 4:
            System.out.print("Enter ID: ");
            manager.deleteDocument(sc.nextInt());
            break;

        case 5:
            System.out.println("Exiting program...");
            return;

        default:
            System.out.println("Invalid choice!");
```

```
            }
        }
    }
}
```

6. INPUT AND OUTPUT SCREENS

```
===== DDRMS MENU =====
1. Add Document
2. View Documents
3. Search Document
4. Delete Document
5. Exit
Enter choice: 1
Enter ID: 101
Enter Title: java notes
Enter Author: Tharaniish
Enter Date: 2026
```

```
===== DDRMS MENU =====
1. Add Document
2. View Documents
3. Search Document
4. Delete Document
5. Exit
Enter choice: 2

ID: 101
Title: Java notes
Author: Tharaniish
Date: 2026
------------------------

ID: 101
Title: java notes
Author: Tharaniish
Date: 2026
------------------------
```

```
------------------------

===== DDRMS MENU =====
1. Add Document
2. View Documents
3. Search Document
4. Delete Document
5. Exit
Enter choice: 3
Enter ID: 101

ID: 101
Title: Java notes
Author: Tharaniish
Date: 2026
```

```
===== DDRMS MENU =====
1. Add Document
2. View Documents
3. Search Document
4. Delete Document
5. Exit
Enter choice: 4
Enter ID: 101
Document deleted.
```

```
===== DDRMS MENU =====
1. Add Document
2. View Documents
3. Search Document
4. Delete Document
5. Exit
Enter choice: 5
Exiting program...
```

7. DISCUSSION & CONCLUSION

7.1 Discussion

The **Document & Digital Record Management System (DDRMS)** was developed to automate the process of managing digital documents in an efficient and organized manner. The system allows users to add documents, store them securely, search records easily, and delete unwanted files.

During the development process, Core Java programming concepts such as loops, conditional statements, file handling, methods, and object-oriented principles were applied. The system successfully performs document storage, retrieval, and management operations without errors.

The system was tested with different inputs to ensure that all functionalities such as adding, searching, viewing, and deleting documents work correctly. The results show that the system reduces manual effort, improves accuracy, and enhances the efficiency of document management.

7.2 Conclusion

The **Document & Digital Record Management System (DDRMS)** demonstrates the practical use of Java programming in managing digital documents effectively. The system provides a simple and efficient way to store, retrieve, and manage records. It reduces the dependency on manual paperwork and minimizes the chances of data loss.

The application successfully implements core Java concepts such as object-oriented programming and file handling. All the functionalities like adding, searching, viewing, and deleting documents work correctly. The system improves accuracy and saves time in document management. Overall, DDRMS is a reliable and user-friendly solution for digital record management.

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com